Exponential online mistakes for tanh networks with stronger readout regularization

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Abstract

A Bayesian neural network can learn a small target from polynomially many IID examples and nevertheless make exponentially many mistakes on one fixed ordering of the input cube. We prove this for ordinary tanh, one hidden layer, independent Gaussian weights with equal variances across all first-layer coordinates, and every sufficiently large hidden width. The prior has first-layer variance $\tau^2 N^{1-\gamma}/d$ and readout variance $\tau^2 N^{-\gamma-1}/A_N$, where $\gamma \geq 1$, $\tau > 0$ are fixed and $A_N \to \infty$ can diverge arbitrarily slowly. The target has a Boolean circuit of size $\Theta(k)$ and a shallow tanh teacher of width $O(k^4)$ with every weight bounded by one. After an $O(k)$ calibration prefix, a fixed input permutation forces at least 2^{k-1} designated mistakes, with posterior wrong-label probability tending to one as $N \to \infty$. The same posterior has polynomial IID sample complexity. The proof gives an exact scalar-optimality theorem: sufficiently strong readout regularization makes the minimum energy of any width equal to an explicit function of the affine margin norm. A Gaussian-volume comparison then handles the growing parameter dimension. A subpolynomial readout correction, such as a logarithmic one, preserves the corresponding power-law exponents. The result does not establish the case $A_N \equiv 1$ or a lower bound at every realizable width.

1 Model and main result

An IID guarantee controls errors when training and testing use the same input distribution. An online mistake bound would be stronger: it would control the total number of errors even when the inputs arrive in an unfavorable order. We exhibit a separation between these guarantees for an exact Bayesian tanh learner. The network and target use the same shallow architecture; the lower bound persists at arbitrarily large learner width.

Let $X = \{-1, 1\}^d$ and $\bar{x} = (1, x) \in \mathbb{R}^D$, where $D = d + 1$. At hidden width N, the learner is

$$F_{u,v}(x) = \sum_{j=1}^{N} v_j \tanh(u_j \cdot \bar{x}). \tag{1}$$

All weights are independent centered Gaussians, with

$$\mathrm{Var}(u_{ji}) = \frac{\tau^2 N^{1-\gamma}}{d}, \qquad \mathrm{Var}(v_j) = \frac{\tau^2 N^{-\gamma-1}}{A_N}. \tag{2}$$

The constant input coordinate has the same first-layer variance as every other coordinate. Fix $\gamma \geq 1$, $\tau > 0$, and any positive sequence $A_N \to \infty$. No monotonicity or minimum rate of divergence is assumed. For example one may use $A_N = \log\log(N + e^e)$, or a slower divergent sequence.

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For a Boolean function g define its strict global-margin cell

$$U_g = \{(u, v) : g(x)F_{u,v}(x) > 1/2 \text{ for every } x \in X\}. \tag{3}$$

The prior is conditioned on the union of these cells. After every prediction the true label is revealed, and the posterior retains exactly the cells that fit all observed labels. The online input order below has no repetitions. The posterior error probability is the mass of functions with the wrong label at the next input. This is also the error probability of a fresh posterior sample; posterior-label majority is wrong when it exceeds $1/2$. The confidence convention requiring more than $2/3$ posterior probability on the correct label is covered as well.

The factor A_N is the substantive prior change in this theorem. The reference normalization has $A_N = 1$, with readout variance $\tau^2 N^{-\gamma-1}$. We prove the lower bound after dividing that variance by any diverging factor. If this factor is subpolynomial, the layerwise power exponents are unchanged. The result does not remove the factor altogether.

Theorem 1. *For every $k \geq 3$, put $d = 2k + 1$ and write inputs as (x, y, z) with $x, y \in \{-1, 1\}^k$ and $z \in \{-1, 1\}$. Define*

$$X_x = \sum_{i=0}^{k-1} 2^i \frac{x_i + 1}{2}, \qquad Y_y = \sum_{i=0}^{k-1} 2^i \frac{y_i + 1}{2}, \qquad f_k(x, y, z) = +1 \iff z = -1 \text{ and } X_x \geq Y_y.$$

There is one fixed permutation of the input cube with the following properties.

1. *The target has a bounded-fan-in Boolean circuit of size $\Theta(k)$. It also has a one-hidden-layer tanh teacher with all weights bounded by one, global output margin greater than $3/4$, and hidden width at most*

$$n_k = \left\lceil 288(k + 2)^2(4k + 3)(2k + 3) \log 2 \right\rceil = O(k^4).$$

2. *The permutation begins with at most $4k+3$ calibration examples and then has $T_k = 2^k - k - 1 \geq 2^{k-1}$ designated rounds. For every fixed choice of $\gamma \geq 1$, $\tau > 0$, and $A_N \to \infty$ in (2), the posterior wrong-label probability tends to one at each designated round as $N \to \infty$.*

3. *Thus at every sufficiently large width, posterior-label majority makes all T_k mistakes and the strict $2/3$ correctness criterion fails at all those rounds. For every $\eta > 0$, posterior sampling makes all T_k mistakes with probability greater than $1 - \eta$ at every sufficiently large width.*

4. *For the same prior and target, $O((d^2 + \log(1/\beta))/\varepsilon)$ IID examples suffice, with probability at least $1 - \beta$, for posterior correct-label probability greater than $2/3$ on input-distribution mass at least $1 - \varepsilon$, at every sufficiently large width. The width threshold can be independent of the training sample and its input distribution.*

The target and permutation depend only on k. The width threshold may depend on k, γ, τ, A_N, and the required confidence. It is not claimed to be polynomial, or to equal the teacher width.

The errors are exponential in the target's Boolean circuit size, and superpolynomial in the displayed bounded-weight shallow teacher bound. No choice among tied minimum-energy predictors is used. The comparison below bounds every network that would make the correct next prediction.

2 Exact scalar optimality at every width

Write $b_j = Nv_j$ and temporarily fix a readout penalty $A > 0$. The Gaussian rate energy is

$$E_A = \frac{d}{N}\sum_{j=1}^{N}\|u_j\|^2 + \frac{A}{N}\sum_{j=1}^{N} b_j^2, \qquad F(x) = \frac{1}{N}\sum_j b_j \tanh(u_j \cdot \bar{x}). \tag{4}$$

The next theorem concerns arbitrary finite linearly separable training sets. Its lower bound does not assume that the network's output sign function is an affine threshold.

Let $z_i = y_i\bar{x}_i \in \mathbb{R}^D$ be the signed data vectors and let a_* minimize $\|a\|^2$ subject to $z_i \cdot a \geq 1$. Put $Q = \|a_*\|^2 > 0$. The minimizer is unique. Its KKT conditions give a set I of active constraints and positive coefficients μ_i such that

$$a_* = \sum_{i\in I}\mu_i z_i, \qquad z_i \cdot a_* = 1 \ (i \in I), \qquad \sum_{i\in I}\mu_i = Q. \tag{5}$$

Set $\rho_i = \mu_i/Q$ and $p = \min\{1/2, \min_{i\in I}\rho_i\} > 0$. Define

$$\Phi_A(Q) = \min_{r>0}\left\{dQr^2 + \frac{A}{4\tanh^2 r}\right\}. \tag{6}$$

Its unique minimizer $r = r_A(Q)$ satisfies

$$A = \frac{4dQr\tanh^3 r}{\operatorname{sech}^2 r}. \tag{7}$$

The right side is strictly increasing from zero to infinity. In particular $r_A(Q)$ increases with A and decreases with Q. For $Q_1 > Q_0$, evaluation at $r_A(Q_1)$ gives

$$\Phi_A(Q_1) - \Phi_A(Q_0) \geq d(Q_1 - Q_0)r_A(Q_1)^2 > 0. \tag{8}$$

Theorem 2 (An exact all-width minimum)**.** *If*

$$A \geq \frac{256dQ\log(16/p)}{p^2}, \tag{9}$$

then, for every $N \geq 1$, the minimum of (4) over all width-N tanh networks satisfying the training output margins $y_iF(x_i) \geq 1/2$ equals $\Phi_A(Q)$. It is attained by cloning the single neuron

$$u_j = r_A(Q)a_*, \qquad b_j = \frac{1}{2\tanh r_A(Q)}.$$

The infimum with strict training margins is the same.

Proof. The displayed clone satisfies every training margin and has energy $\Phi_A(Q)$. We prove a matching lower bound for every network. Write $r = r_A(Q)$, and set

$$\lambda = \frac{A}{\tanh^2 r}, \qquad K = \frac{\lambda^2}{4A}, \qquad S(u) = \sum_{i\in I}\rho_i \tanh(z_i \cdot u).$$

For any scalar b, completing the square gives

$$d\|u\|^2 + Ab^2 - \lambda bS(u) \geq d\|u\|^2 - KS(u)^2. \tag{10}$$

Oddness permits replacing u by $-u$ when necessary, so we may suppose $S(u) \geq 0$ when bounding the right side.

If every active score $z_i \cdot u$ is nonnegative, concavity of tanh on $[0, \infty)$, (5), and Cauchy–Schwarz imply

$$S(u) \leq \tanh\left(\sum_i \rho_i z_i \cdot u\right) = \tanh(a_* \cdot u/Q) \leq \tanh(\|u\|/\sqrt{Q}). \tag{11}$$

Thus (10) is at least

$$\min_{s \geq 0}\{dQs^2 - K\tanh^2 s\} = dQr^2 - K\tanh^2 r =: \zeta. \tag{12}$$

To check the equality, $h(v) = \tanh^2\sqrt{v}$ is strictly concave for $v > 0$. Put $s = \sqrt{v}$. Then $h'(v) = \tanh(s)\operatorname{sech}^2(s)/s$. The logarithmic derivative of this expression with respect to s is

$$\frac{1}{\sinh s \cosh s} - 2\tanh s - \frac{1}{s} < 0.$$

Here $\sinh s \cosh s > s$ for $s > 0$. Consequently the objective in (12) is strictly convex as a function of s^2, and (7) makes $s = r$ stationary.

If instead one active score is negative, that term contributes at most zero while all others contribute at most their weights. Hence $S(u) \leq 1 - p$, and (10) is at least $-K(1-p)^2$. This is at least ζ provided

$$\tanh^2 r \geq 1 - p, \qquad Ap/8 \geq dQr^2. \tag{13}$$

Indeed $\tanh^2 r - (1-p)^2 \geq p(1-p) \geq p/2$ and $K \geq A/4$.

We verify (13) from (9). Put $r_0 = \log(16/p)$. Since $\sinh^3 r/\cosh r \leq e^{2r}/4$, the right side of (7) at r_0 is at most $256dQ\log(16/p)/p^2$. Thus $r \geq r_0$. Then $\operatorname{sech}^2 r \leq 4e^{-2r} \leq p^2/64 \leq p$. Also $\tanh r \geq 1/2$, so (7) gives

$$\frac{Ap}{8dQr^2} = \frac{p\tanh^3 r}{2r\operatorname{sech}^2 r} \geq \frac{pe^{2r}}{64r} \geq \frac{4}{p\log(16/p)} > 1.$$

The middle inequality uses that e^{2r}/r increases for $r \geq r_0$; the last uses $p\log(16/p) \leq \frac{1}{2}\log 32 < 4$.

We have proved the same atomwise lower bound ζ in both cases. Average (10) over the neurons. The training constraints imply $N^{-1}\sum_j b_j S(u_j) \geq 1/2$. Therefore

$$E_A \geq \lambda/2 + \zeta = dQr^2 + \frac{A}{4\tanh^2 r} = \Phi_A(Q).$$

Finally, an arbitrarily small readout amplification makes the cloned training margins strict and approaches the same energy. □

Corollary 3 (A common Boolean-data threshold)**.** *Suppose $z_i \in \{-1, 1\}^D$ and $Q \leq Q_{\max}$, where $Q_{\max} \geq 1$. For every such training set, Theorem 2 holds whenever*

$$A \geq A_0 := 1024dQ_{\max}^3 D^{2D}\log(32Q_{\max}D^D). \tag{14}$$

This bound is independent of the number of training examples and of N.

Proof. Choose a KKT certificate with minimal positive support. Its active vectors are linearly independent: otherwise a linear dependence lets one vary the nonnegative coefficients until one vanishes, preserving their conic combination. Thus the support size is $r \leq D$. Its Gram matrix

G is an integer positive-definite matrix with diagonal D, and $G\mu = \mathbf{1}$. Hadamard's inequality gives $1 \le \det G \le D^r \le D^D$. Cramer's rule and positivity imply $\mu_i \ge 1/\det G \ge D^{-D}$, since each numerator is a positive integer. Hence $\rho_i \ge 1/(Q_{\max} D^D)$. The common value $p_0 = 1/(2Q_{\max} D^D)$ is a valid smaller bound for p. Substituting it and $Q_{\max}$ into (9) gives (14). □

For $D = O(k)$ and $Q_{\max} = \exp(O(k))$, this sufficient penalty is $\exp(O(k \log k))$. The theorem nevertheless allows any divergent sequence A_N: it eventually exceeds the fixed threshold associated with each fixed target. A slow choice can make the required width large.

3 A fixed input order with exponentially many affine mistakes

We give the complete affine construction used by the network theorem. Keep the target f_k from Theorem 1, and abbreviate the two encoded integers by X and Y. The calibration examples all have selector $z = -1$.

1. Take diagonal inputs $x = y$ equal to the all-positive vector, each of its k one-coordinate flips, and the negatives of these vectors. Every label is positive.

2. For each $i \in \{0, \dots, k-1\}$, take two negative examples with $x_i = -1, y_i = +1$ and $x_j = +1, y_j = -1$ for $j < i$. For $j > i$, set the equal bits $x_j = y_j$ all positive in one example and all negative in the other. Their highest differing bit is i, so $X < Y$.

Remove duplicate examples. The resulting calibration has at most $4k + 2$ inputs. Let q_t denote the input $z = +1$, $Y = 0$, $X = t$; its target label is negative for every $0 \le t \le 2^k - 1$.

Lemma 4. *For $k \le t \le 2^k - 1$, the unique minimum-norm affine unit-margin coefficient vector fitting the calibration and the negative example q_t is the vector of the affine score*

$$a_t \cdot \bar{x} = -t - (t+1)z + \sum_{i=0}^{k-1} 2^i (x_i - y_i). \tag{15}$$

It fits all earlier q_j with $j \le t$, and its absolute score is at least one on the entire cube. Its squared norm is

$$Q_t = t^2 + (t+1)^2 + 2\sum_{i=0}^{k-1} 4^i. \tag{16}$$

For $t < 2^k - 1$, it predicts q_{t+1} positively, and adding the correct negative label raises the convex affine minimum by

$$Q_{t+1} - Q_t = 4(t+1) > 0. \tag{17}$$

Proof. Write an arbitrary affine score as

$$b + cz + \sum_i a_i x_i + \sum_i h_i y_i,$$

and set

$$u = b - c, \quad v = b + c, \quad w_i = (a_i - h_i)/2, \quad s_i = a_i + h_i, \quad U = u - 1.$$

The squared coefficient norm is

$$\frac{u^2 + v^2}{2} + 2\sum_i w_i^2 + \frac{1}{2}\sum_i s_i^2. \tag{18}$$

The paired positive diagonal constraints give

$$U \geq 0, \qquad \left|\sum_j s_j\right| \leq U, \qquad \left|\sum_j s_j - 2s_i\right| \leq U.$$

Subtracting the expressions inside the last two absolute values shows $|s_i| \leq U$, hence $\sum_i |s_i| \leq kU$. Average the two negative carry constraints at coordinate i. The upper equal-bit terms cancel and leave

$$u - 2w_i + 2\sum_{j<i} w_j \leq -1.$$

Induction yields

$$w_i \geq \sum_{j<i} w_j + \frac{u+1}{2} \geq 2^i(1 + U/2). \tag{19}$$

At q_t, the non-selector part of the input contribution is $2\sum_{\mathrm{bit}_i(t)=1} w_i - \sum_{\mathrm{bit}_i(t)=0} s_i$. Its negative unit-margin constraint therefore implies

$$v \leq -1 - 2\sum_{\mathrm{bit}_i(t)=1} w_i + \sum_{\mathrm{bit}_i(t)=0} s_i \leq -(2t+1) + (k-t)U \leq -(2t+1),$$

since $t \geq k$. Combining this with $u \geq 1$ and (19) in (18) gives

$$\|a\|^2 \geq \frac{1 + (2t+1)^2}{2} + 2\sum_i 4^i = Q_t.$$

Equality requires $u = 1$, $s_i = 0$, $w_i = 2^i$ and $v = -(2t+1)$, which is exactly (15).

Conversely the displayed score equals $1 + 2(X - Y)$ on $z = -1$, and $-(2t+1) + 2(X - Y)$ on $z = +1$. It fits every calibration label, and its score at q_j is $2j - 2t - 1 \leq -1$ for $j \leq t$. Both branch scores are odd integers on the full cube, so the absolute margin is at least one. At q_{t+1} the score is $+1$. Applying the same minimum calculation to the augmented history gives Q_{t+1}, and subtraction gives (17). □

The input permutation starts with the calibration, then q_k, then $q_{k+1}, \ldots, q_{2^k-1}$, and finally every remaining input in lexicographic order. These blocks are disjoint. Its calibration prefix has length at most $4k + 3$, and there are $T_k = 2^k - k - 1 \geq 2^{k-1}$ designated queries. Just before each one, the wrong affine witness has the exact convex margin minimum for all data seen so far. The upper bound $Q_t \leq 3 \cdot 4^k$ suffices in Corollary 3.

The target has a linear-size Boolean circuit. With 0/1 bit notation, comparison of the lowest $i + 1$ bits obeys

$$G_{-1} = 1, \qquad G_i = (X_i \wedge \neg Y_i) \ \vee \ ((X_i \leftrightarrow Y_i) \wedge G_{i-1}).$$

Each step uses a constant number of bounded-fan-in gates; conjoin the last result with $z = -1$. Every input bit is essential: for x_i use $Y = 2^i$ and all other X bits zero; for y_i use $X = 0$ and all other Y bits zero; for z take $X = Y$. A fan-in-two circuit with d essential inputs has at least $d - 1$ gates, by counting edges in the connected ancestry of its output. This proves the matching $\Theta(k)$ bound.

4 Bounded-weight shallow teachers from Fourier coefficients

The target used above has a small teacher at the same shallow depth as the learner. We prove this using a general construction that may also be useful for other targets. All networks in this section have one tanh hidden layer, one constant input coordinate, and a linear readout.

For $S \subseteq \{1, \dots, d\}$, write $\chi_S(x) = \prod_{i \in S} x_i$. A Boolean function has the Fourier expansion

$$f(x) = \sum_{S \subseteq \{1,\dots,d\}} \widehat{f}(S)\chi_S(x), \qquad \mathcal{A}(f) = \sum_S |\widehat{f}(S)|.$$

The Fourier coefficients here use the uniform measure on the Boolean cube; the learning theorem does not require a uniform input distribution.

Theorem 5 (A bounded-weight shallow teacher)**.** *Every $f : \{-1,+1\}^d \to \{-1,+1\}$ has a one-hidden-layer tanh implementation that uniformly approximates f within $1/4$, has every weight of absolute value at most 1, and has hidden width at most*

$$\left\lceil 288\mathcal{A}(f)^2(2d+1)(d+2)\log 2 \right\rceil. \tag{20}$$

In particular, its signed output is greater than $3/4$ at every input.

Lemma 6 (Parity correlation of bounded tanh features)**.** *For $r \geq 0$, let $b_r = 0$ when r is odd and $b_r = 1$ when r is even. For independent uniform signs $s_1, \dots, s_r$, define*

$$c_r = \mathbb{E}_s\left[\left(\prod_{i=1}^{r} s_i\right)\tanh\left(\sum_{i=1}^{r} s_i + b_r\right)\right].$$

Then

$$|c_r| \geq \frac{1}{3\sqrt{2r+1}}. \tag{21}$$

The empty product and sum give $c_0 = \tanh 1$.

Proof. Put $m = \lfloor r/2 \rfloor$ and $\beta_m = \binom{2m}{m}/4^m$. The highest-degree Fourier coefficient of $\operatorname{sign}(\sum_i s_i + b_r)$ is $(-1)^m\beta_m$. For $r \geq 1$, this follows by splitting the binomial sum at $\lfloor r/2 \rfloor$ and using

$$\sum_{j=0}^{q}(-1)^j\binom{r}{j} = (-1)^q\binom{r-1}{q};$$

the case $r = 0$ is immediate.

Every possible value of $T = \sum_i s_i + b_r$ is an odd integer, and each individual value has probability at most β_m. For even r this is the central-binomial bound. For odd $r = 2m+1$, the largest probability is $\frac{2m+1}{2m+2}\beta_m$. Since $|\operatorname{sign}(t) - \tanh t| \leq 2e^{-2|t|}$ for $t \neq 0$,

$$\begin{aligned} |c_r - (-1)^m\beta_m| &\leq \mathbb{E}\,|\operatorname{sign}(T) - \tanh T| \\ &\leq 4\beta_m\sum_{j\geq 0} e^{-2(2j+1)} = \frac{4e^{-2}}{1-e^{-4}}\beta_m < \frac{7}{12}\beta_m. \end{aligned}$$

The last inequality follows from $e^2 > 7$ and $4e^{-2}/(1-e^{-4}) = 4e^2/(e^4-1) < 7/12$. Thus $|c_r| > \frac{5}{12}\beta_m > \frac{1}{3}\beta_m$.

Finally, $\beta_m \geq 1/\sqrt{4m+1}$. This follows by induction from $\beta_0 = 1$ and $\beta_{m+1}/\beta_m = (2m+1)/(2m+2)$: after squaring, the induction step is

$$(2m+1)^2(4m+5) \geq 4(m+1)^2(4m+1),$$

whose left-hand side exceeds the right-hand side by 1. Since $4m+1 \leq 2r+1$, the claimed bound follows. □

Proof of Theorem 5. For a subset S of cardinality r and a sign vector s indexed by S, let

$$h_{S,s}(x) = \tanh\left(\sum_{i\in S} s_i x_i + b_r\right).$$

Every affine coefficient belongs to $\{0, -1, +1\}$, including the coefficient of the constant input. Changing variables from s_i to $s_i x_i$ gives the exact identity

$$\mathbb{E}_s\left[\left(\prod_{i\in S} s_i\right) h_{S,s}(x)\right] = c_r \chi_S(x). \tag{22}$$

Consequently, the Fourier expansion expresses f as

$$f(x) = B\mathbb{E}[H(x)], \qquad B = \sum_S \frac{|\widehat{f}(S)|}{|c_{|S|}|} \leq 3\mathcal{A}(f)\sqrt{2d+1},$$

where H is a signed feature of the form $h_{S,s}$ and $|H(x)| \leq 1$. Explicitly, choose S with probability proportional to $|\widehat{f}(S)|/|c_{|S|}|$, choose s uniformly, and multiply $h_{S,s}$ by the signs prescribed by $\widehat{f}(S)/c_{|S|}$ and $\prod_{i\in S} s_i$. The empty subset uses the constant feature $\tanh 1$ and needs no separate construction.

Take

$$n = \left\lceil 32B^2 \log(4 \cdot 2^d)\right\rceil$$

independent features $H_1, \ldots, H_n$. Hoeffding's inequality and a union bound over the cube give

$$\mathbb{P}\left\{\max_x \left|\frac{B}{n}\sum_{j=1}^{n} H_j(x) - f(x)\right| \geq \frac{1}{4}\right\} \leq 2 \cdot 2^d \exp\left(-\frac{n}{32B^2}\right) \leq \frac{1}{2}.$$

A choice of features therefore achieves uniform error strictly below $1/4$. Put each feature's sign in its readout coefficient, whose magnitude is B/n. Because $\mathcal{A}(f) \geq 1$ and $|c_r| \leq 1$, we have $B \geq 1$ and $n \geq B$; hence every readout weight also has magnitude at most 1. Finally, $B^2 \leq 9\mathcal{A}(f)^2(2d+1)$ and $\log(4 \cdot 2^d) = (d+2)\log 2$ give (20). □

Corollary 7 (A shallow teacher for the selector target)**.** *For the target f_k on $d = 2k+1$ inputs, equal to $+1$ precisely when $z = -1$ and $X \geq Y$, there is a one-hidden-layer tanh teacher with all weights bounded by 1, global signed output greater than $3/4$, and width at most*

$$W_k = \left\lceil 288(k+2)^2(4k+3)(2k+3)\log 2\right\rceil = O(k^4). \tag{23}$$

In the convention that C_2 bounds both hidden width and individual weight magnitudes, $C_2(f_k) \leq W_k$.

Proof. Let x_i, y_i be the two integers' Boolean bits, with $i = 0$ the least significant position, and put $E_i = (1 + x_i y_i)/2$. The comparison sign G, equal to $+1$ when $X \geq Y$ and -1 otherwise, satisfies

$$G(x, y) = \prod_{i=0}^{k-1} E_i + \sum_{i=0}^{k-1} \frac{x_i - y_i}{2} \prod_{j=i+1}^{k-1} E_j.$$

The first term handles equality; otherwise the highest differing bit contributes the comparison sign and all other summands vanish. Each E_i and $(x_i - y_i)/2$ has Fourier ℓ^1 norm 1. Submultiplicativity and the triangle inequality therefore give $\mathcal{A}(G) \leq k + 1$. The selector target is

$$f_k(x, y, z) = \frac{1 - z}{2} G(x, y) - \frac{1 + z}{2},$$

so $\mathcal{A}(f_k) \leq k + 2$. Apply Theorem 5 with $d = 2k + 1$. □

5 From energy separation to the actual posterior

We now control the posterior at a prescribed width-dependent variance, rather than first sending the variance to zero separately at each width. The correction A_N may tend to infinity arbitrarily slowly. Throughout this section $d \geq 1$ is fixed, $D = d + 1$, and

$$\sigma_N^2 = \tau^2 N^{-\gamma}, \qquad \gamma \geq 1, \quad \tau > 0, \quad A_N \longrightarrow \infty.$$

All constants in the estimates may depend on the fixed dimension, the fixed finite training histories, γ, and τ, but not on N.

Proposition 8 (A direct posterior comparison)**.** *Let $\mathcal{S}$ be a fixed finite labeled history. Suppose its affine unit-margin problem has minimum-norm solution a_0, with $Q_0 = \|a_0\|^2 > 0$, and*

$$|a_0 \cdot \bar{x}| \geq 1 \qquad \text{for every } x \in \{-1, 1\}^d.$$

Let (x_*, y_*) be a further labeled point such that $y_* a_0 \cdot \bar{x}_* < 0$. Suppose the affine unit-margin problem for $\mathcal{S} \cup \{(x_*, y_*)\}$ is feasible, with squared minimum norm $Q_1 > Q_0$. Under the prior (2), conditioned on the global absolute output margin greater than $1/2$ and on the exact labels in $\mathcal{S}$, the posterior probability of label y_* at x_* tends to zero as $N \to \infty$. The conclusion holds simultaneously for any fixed finite collection of such histories and next points.

Proof. Write $A = A_N$ and $\sigma = \sigma_N$. At every sufficiently large width, Theorem 2 applies to both affine problems, because their required penalties are fixed and $A_N \to \infty$. Put

$$E_0 = \Phi_A(Q_0), \qquad E_1 = \Phi_A(Q_1), \qquad \Delta = Q_1 - Q_0 > 0.$$

The old problem has a cloned scalar witness of energy E_0, with the same full-cube sign function as a_0. Every network fitting the old history and the additional correct label, with output margins at least $1/2$ there, has energy at least E_1. The latter assertion includes all nonlinear output functions.

Let $r_A(Q)$ minimize the scalar profile (6). Its stationarity equation is

$$A = 4dQ\, r_A(Q) \frac{\tanh^3 r_A(Q)}{\operatorname{sech}^2 r_A(Q)}.$$

The right side increases strictly with its radius. At $r = (\log A)/4$ it is at most $dQre^{2r}$, which is smaller than A for all sufficiently large A. Thus $r_A(Q_1) \geq (\log A)/4$ eventually. Evaluating the profile for Q_0 at the minimizer for Q_1 gives

$$E_1 - E_0 \geq d\Delta\, r_A(Q_1)^2 \geq c(\log A)^2, \qquad c = d\Delta/16 > 0. \tag{24}$$

Also, for $i = 0, 1$,

$$E_i = A/4 + O((\log A)^2), \tag{25}$$

because the readout contribution is at least $A/4$, and the cloned choice $r = (\log A)/2$ gives the matching upper estimate.

A strict-margin ball. Use the normalized readout variables $b_j = Nv_j$ and whiten the energy by

$$U_j = \sqrt{d/N}\, u_j, \qquad B_j = \sqrt{A/N}\, b_j.$$

There are $P = N(d+2)$ coordinates, each independently distributed as $N(0, \sigma^2)$, and their squared Euclidean norm is E_A. Set

$$G = \frac{c}{2}(\log A)^2, \qquad \varepsilon_N = \frac{G}{16E_0}.$$

Amplify the scalar witness's readout by $1 + \varepsilon_N$. For large N, $\varepsilon_N < 1$, its center energy is at most $E_0 + 3G/16 < E_0 + G/4$, and its signed output on every cube input is at least $(1 + \varepsilon_N)/2$. Its global margin slack is therefore $\varepsilon_N/2$, where

$$\varepsilon_N = \Theta((\log A)^2/A).$$

In whitened coordinates the output is

$$F(x) = \frac{1}{\sqrt{AN}} \sum_{j=1}^{N} B_j \tanh\left(\sqrt{N/d}\, U_j \cdot \bar{x}\right).$$

Since $|\tanh| \leq 1$ and $|\tanh'| \leq 1$,

$$\|\nabla F(x)\| \leq A^{-1/2} + \sqrt{D/(Ad)}\, \|B\|.$$

On the radius-one ball about the amplified center, this is bounded by a fixed constant, uniformly over all cube inputs: the center energy is $O(A)$. Consequently the Euclidean ball of radius

$$r_N = A^{-2}$$

lies inside the same strict global-margin sign cell for every sufficiently large N. Indeed its output perturbations are $O(A^{-2})$, whereas the available slack is of order $(\log A)^2/A$. Its maximum energy is at most $E_0 + G/2$, because the additional energy is bounded by $2\sqrt{E_{\text{center}}}\, r_N + r_N^2 = O(A^{-3/2}) = o(G)$.

The cube of side $2r_N/\sqrt{P}$ centered at this witness is contained in the ball. Integrating the minimum Gaussian density on the cube gives

$$\begin{aligned} \log \mathbb{P}(\text{witness cell}) &\geq -\frac{E_0 + G/2}{2\sigma^2} + P\log 2 - 2P\log A - \frac{P}{2}\log(2\pi P\sigma^2) \\ &\geq -\frac{E_0 + G/2}{2\sigma^2} - C_1 N(1 + \log A). \end{aligned} \tag{26}$$

The last bound uses $P\sigma^2 = (d+2)\tau^2 N^{1-\gamma} \leq (d+2)\tau^2$. Keeping this factor together is essential when A_N diverges slowly.

All correct-label competitors. Their prior event is contained in $\{E_A \geq E_1\}$. Since E_A/σ^2 has the chi-square distribution with P degrees of freedom, the exponential-moment bound gives, whenever $E_1 > P\sigma^2$,

$$\begin{aligned}\log \mathbb{P}(\text{correct-label event}) &\leq -\frac{E_1}{2\sigma^2} + \frac{P}{2}\left[1 + \log \frac{E_1}{P\sigma^2}\right] \\ &\leq -\frac{E_1}{2\sigma^2} + C_2 N[1 + \log A + (\gamma - 1)\log N]. \end{aligned} \tag{27}$$

The condition holds eventually by (25) and $A_N \to \infty$. The posterior denominator contains the witness cell. Since $E_1 - E_0 \geq 2G$, dividing (27) by (26) yields

$$\begin{aligned}\log \mathbb{P}(y_* \text{ at } x_* \mid \mathcal{S},\ \text{global margin}) &\leq -\frac{3G}{4\sigma^2} + C_3 N[1 + \log A + (\gamma - 1)\log N] \\ &= -c' N^{\gamma} (\log A_N)^2 + C_3 N[1 + \log A_N + (\gamma - 1)\log N] \end{aligned} \tag{28}$$

for a fixed $c' > 0$. At $\gamma = 1$, the negative term dominates because $\log A_N \to \infty$. At $\gamma > 1$, it also dominates $N \log N$, since $N^{\gamma-1}(\log A_N)^2 / \log N \to \infty$. This proves the asserted convergence. A finite collection of histories allows a common width threshold. □

6 Polynomial IID learning under the same prior

The preceding lower bound concerns an adversarial input order. The same prior has a polynomial realizable IID guarantee for every affine threshold target, including the target used in the online construction.

Lemma 9 (A gap for nonthreshold output functions)**.** *Let $M = 2^d$. If a Boolean function g is not an affine threshold function, then every width-N tanh network with $g(x)F(x) \geq 1/2$ at all cube inputs has*

$$E_A \geq \frac{AM^2}{4(M-1)^2}.$$

Proof. Let $\mathbb{E}_j$ denote uniform averaging over neurons and put $C = \mathbb{E}_j |b_j|$. For every input x, the margin constraint gives

$$\mathbb{E}_j \left[|b_j| \{1 - \operatorname{sign}(b_j) g(x) \tanh(u_j \cdot \bar{x})\}\right] \leq C - 1/2.$$

Assign either sign when $b_j = 0$. For each neuron there is some input at which $\operatorname{sign}(b_j) g(x) u_j \cdot \bar{x} \leq 0$; otherwise that oriented affine score would represent g strictly on the whole cube. Thus at least one of its bracketed deficits is at least one, and every deficit is nonnegative. Summing the displayed inequalities over all M inputs gives $C \leq M(C - 1/2)$, hence $C \geq M/[2(M-1)]$. Finally $E_A \geq A\mathbb{E}_j b_j^2 \geq AC^2$. □

Theorem 10 (IID sample complexity)**.** *Fix an affine threshold target $f : \{-1,1\}^d \to \{-1,1\}$, any input distribution $\mathcal{D}$, and $\epsilon, \beta \in (0,1)$. If*

$$m \geq \frac{4}{\epsilon}\left[(d+1)^2 \log 2 + \log(1/\beta)\right],$$

then, with probability at least $1 - \beta$ over m IID examples labeled by f, the posterior under (2) assigns probability greater than $2/3$ to the correct label on a set of $\mathcal{D}$-probability at least $1 - \epsilon$, at every sufficiently large width. In particular posterior-label majority has population error at most ϵ. The width threshold may depend on $f, d, \epsilon, \gamma, \tau$ and the sequence A_N, but can be chosen independently of the training sample and $\mathcal{D}$.

Proof. Choose a full-cube affine unit-margin implementation of f with squared norm $Q_f < \infty$. Cloning it with scalar radius $(\log A)/2$ gives energy $A/4 + O((\log A)^2)$. The same small readout amplification and radius-A^{-2} whitened ball used above give a strict full-target cell whose prior mass is at least

$$\exp\left[-\frac{A/4 + O((\log A)^2)}{2\sigma_N^2} - CN(1 + \log A)\right].$$

This cell is consistent with every sample labeled by f. By Lemma 9, the union of all nonthreshold output cells has energy at least

$$\frac{AM^2}{4(M-1)^2} \geq \frac{A}{4} + \frac{A}{2M}.$$

Since $(\log A)^2 = o(A)$, applying the same chi-square bound and comparing with the full-target cell shows that the posterior mass of all nonthreshold functions is at most a quantity $B_N \to 0$, uniformly over every target-consistent history. Explicitly, for large N one may take

$$B_N = \min\left\{1, \exp\left[-\frac{A_N}{8M\sigma_N^2} + C'N\{1 + \log A_N + (\gamma - 1)\log N\}\right]\right\}.$$

The exponent tends to minus infinity for every diverging A_N and $\gamma \geq 1$.

Let $\mathcal{H}_d$ be the affine threshold functions on the cube. The 2^d input hyperplanes in coefficient space give the standard region bound

$$|\mathcal{H}_d| \leq \sum_{j=0}^{d+1} \binom{2^d}{j} \leq (2^d + 1)^{d+1} \leq 2^{(d+1)^2}.$$

A realizable consistency union bound therefore shows that, except with probability at most $|\mathcal{H}_d| e^{-m\epsilon/4} \leq \beta$, every threshold function consistent with the sample has population error at most $\epsilon/4$. On this sample event, the posterior mean population error is at most $\epsilon/4 + B_N$. Once $B_N \leq \epsilon/12$, it is at most $\epsilon/3$. At an input where the posterior probability of the correct label is at most $2/3$, the posterior wrong-label probability is at least $1/3$. Markov's inequality thus bounds the probability of these inputs by ϵ. This also bounds majority error, and proves the theorem. □

7 Proof of the main theorem and scope

The fixed calibration and input order in Lemma 4 give, just before each designated query, a convex affine margin minimum Q_t whose optimizer has absolute margin at least one on the entire cube and predicts the next label incorrectly. Adding the correct next label raises the affine minimum to $Q_{t+1} > Q_t$. The exact all-width theorem 2 applies eventually because $A_N \to \infty$. The posterior comparison in the preceding section therefore sends the correct-label probability to zero at each step. There are finitely many steps for each fixed k, so one width threshold suffices for all of them. Using a per-round probability allowance η/T_k and a union bound proves the posterior-sampling assertion without any independence assumption about predictions. The teacher construction and the IID result prove the remaining statements.

Proposition 11 (The scalar witnesses are beaten at the reference scaling)**.** *At $A = 1$, for every $N \geq 2n_k$, the true target has a strict-margin implementation of energy at most*

$$d(d+1) + 18(k+2)^2(2d+1) = O(k^3).$$

Every cloned scalar implementation of the affine witness in Lemma 4 has energy at least

$$\sqrt{dQ_t} \geq \sqrt{\frac{2d}{3}(4^k - 1)}.$$

Thus, for all sufficiently large k, the true target beats every one of these scalar witnesses at every such width under the reference scaling.

Proof. The construction in Theorem 5 supplies a teacher with $n \leq n_k$ neurons, first-row squared norms at most $d + 1$, and readout coefficients of magnitude B/n, where $B \leq 3(k + 2)\sqrt{2d + 1}$. Allocate m_j clones to neuron j, with $\sum_j m_j = N$ and each m_j either $\lfloor N/n \rfloor$ or $\lceil N/n \rceil$. Divide its readout coefficient by m_j. The function is unchanged. Since $N \geq 2n$, each $m_j \geq N/(2n)$, so its canonical energy is at most

$$\frac{d}{N} \sum_j m_j(d + 1) + N \sum_j \frac{(B/n)^2}{m_j} \leq d(d + 1) + 2B^2.$$

For a scalar witness, $\tanh r \leq r$ gives

$$\Phi_1(Q_t) \geq \min_{r>0} \left(dQ_t r^2 + \frac{1}{4r^2} \right) = \sqrt{dQ_t}.$$

The claimed bound on Q_t follows from (16). The exponential lower bound eventually exceeds the polynomial upper bound. □

This comparison does not determine which wide wrong functions may still have low energy. It shows why copying the scalar affine witnesses is not itself a proof for $A = 1$.

The learner uses fixed ordinary tanh, the full Boolean cube, a constant input coordinate, and a Gaussian prior with exact first-coordinate isotropy. The argument includes nonlinear competitors at arbitrary width. The readout correction $A_N \to \infty$ remains essential to this proof: it makes the minimum-energy hidden units saturate on the active training constraints, so affine margin geometry determines the optimum. The argument gives no conclusion for the reference case $A_N \equiv 1$, nor for a fixed bounded readout correction uniformly across this growing target family. Those are separate questions. The theorem is a statistical statement about an exact posterior; it supplies no efficient posterior-sampling algorithm.